Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[aspectratio=1610]beamer
[english]babel
[utf8]inputenc ucs graphicx listings color pgfpages
[FEKT]themevutlecture

Lecture 1: Template for lecture presentation using official colors of Brno University of Technology

Course name/code

John DOE

john@doe.edu

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2019/20

Lists

Unordered list

- Lorem ipsum dolor sit amet, consectetuer adipiscing elit.
- Etiam sapien elit, consequat eget, tristique non, venenatis quis, ante.

Ordered list

1. Lorem ipsum dolor sit amet, consectetuer adipiscing elit.
2. Aliquam erat volutpat:
   - Integer lacinia.
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Multiple columns

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Blocks

Example 1

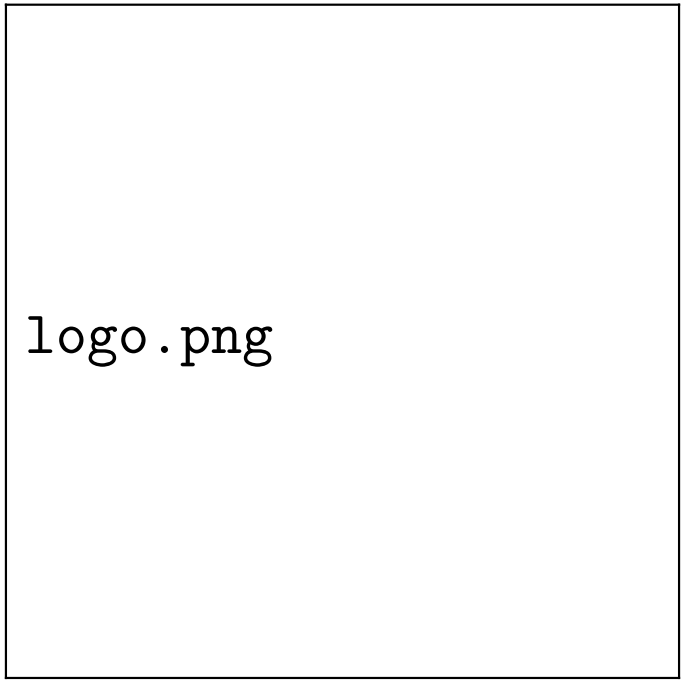
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Solution 1

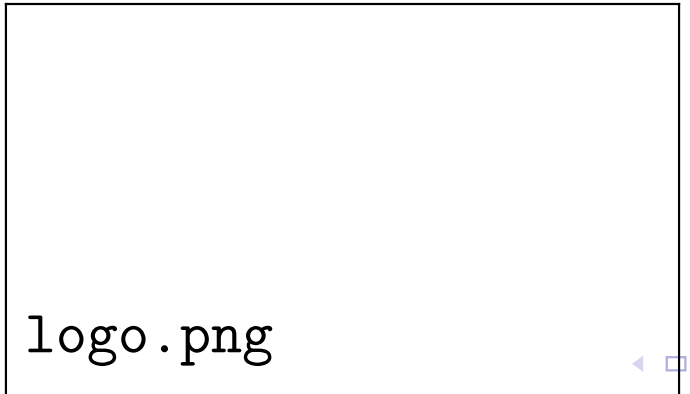
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Figure

language=[LaTeX]TeX



Gambar: Your caption



Table

language=[LaTeX]TeX

Tabel: Your caption

ID	Duration	Complexity	Score
Algo 1	0.0159	0.50	78
Algo 2	0.0453	0.65	88
Algo 3	0.8642	0.77	95

Tabel: Your caption

ID	Duration	Complexity	Score
Algo 1	0.0159	0.50	78
Algo 2	0.0453	0.65	88
Algo 3	0.8642	0.77	95

Equations

Pythagorean theorem can be written as: $a^2 + b^2 = c^2$ where c is the longest side of the triangle, a and b are the other two sides.

Other useful equations (thank you *John Napier*):

language=[LaTeX]TeX

$$\log_b(x^p) = p \cdot \log_b(x) \tag{1}$$

$$\log_b(x^p) = p \cdot \log_b(x) \tag{2}$$

language=[LaTeX]TeX

$\log_b(x) = y$ exactly if $b^y = x$

$\log_b(x) = y$ exactly if $b^y = x$

Code listings

[language=C,title=This is title] void setup(void)
$uart_init(UART_BAUD_SELECT(UART_BAUD_RATE, F_CPU)); //UARTmod$
Fimodule
[language=vhdl,title=This is title,numbers=left]
———————————————————————— – Entity
declaration for hexadecimal to seven-segment decoder
———————————————————————— entity
$hex_to_7segisport(hex_i : instd_logic_vector(4 - 1downto0); seg_o :$
$outstd_logic_vector(7 - 1downto0)); endentityhex_to_7seg;$
[language=Matlab,caption=This is caption] x = 0:0.05:5; y =
$\sin(x.^2); figureplot(x, y)$

Further reading I

[1] MikroElektronika d.o.o. (2019).
PIC Microcontrollers - Programming in C. Online; accessed 8 January 2020.
https://www.mikroe.com/ebooks/pic-microcontrollers-programming-in-c/additional-components.

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Introduction to Giving Presentations.
Klein-Verlag, 1990.

[3] A. Author
On this and that.
Journal of This and That, 2(1):50–100, 2000.